

Constantinos Menelaou

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SHORT RESUMÉ

Sep. 2025–Current	Research Assistant UNIVERSITY COLLEGE DUBLIN · Ireland  Continuation of the PhD work by studying more cases of boulder transport observed in nature.
2021–2026	PhD Candidate UNIVERSITY COLLEGE DUBLIN · Ireland  PhD in Applied Mathematics using Computational Fluid Dynamics to model wave impact on boulders.
2015–2020	Mechanical Engineering with Integrated Masters ARISTOTLE UNIVERSITY OF THESSALONIKI · Greece  5 Year Degree with integrated masters. Specialisation in Fluid Mechanics, Aerodynamics, and Aircraft Design. Thesis on Hypersonic Waverider Aircraft Design.

DEGREES

2025	PhD in Applied Mathematics PHD · UCD 
2020	Mechanical Engineer INTEGRATED MASTERS (BENG & MENG) · AUTH 

CERTIFICATES & GRANTS

2022	UCD Drone, Data Processing and Visualization Course
2021	MaREI PhD Funding
2021	CFD-FEA Course (A Hands-on Introduction to Engineering Simulations)
2021	English IETLS 2021

TEACHING

UCD Tutor	ICHEC High Performance Computing
UCD Tutor	Numerical Algorithms
UCD Tutor	Statistics with Python
UCD Tutor	Mathematical Fluid Mechanics I
UCD Tutor	Analytical Mechanics
UCD Tutor	PDE for Financial Maths

PROGRAMMING

★★★★★ Python, C/C++, Bash, Linux, Conda, HPC
☆☆★★★★ CUDA, MPI, OpenMP, Rust
☆☆★★★★ MATLAB

I have gained experience reading and modifying C++ and CUDA code during my PhD (SPHinXsys and DualSPHysics solvers). Moreover, I am comfortable with OpenFOAM, both the source code and the solver itself. I am familiar with Windows and Linux (Ubuntu and Arch) for development, with my personal preference being Linux. I have also been using the HPC clusters ICHEC's Key, UCD's Sonic and LuxProvide's Meluxina over the years to run my numerical simulations. Furthermore, I am proficient in Python. I have used it for the creation of several tools and packages relevant to my PhD and most of the data analysis. I have also used MATLAB in the past (undergraduate and PhD) whenever Python or other languages were insufficient. Lastly, I am a fan of Rust, which I have used for personal projects such as a gym booking bot with features like a GUI and sound effects.

PUBLICATIONS

- 2026 Displacement and deposition of cliff-bottom rocks using Smoothed Particle Hydrodynamics, *Physics of Fluids* (Accepted, In Press). **Editor's Pick**.
→ Featured in *AIP Scilight*: "Cracking the boulder-tossing mystery" (2026, In Press).
- 2026 Prediction of the displacement by waves of large rocks using smoothed-particle hydrodynamics, *Computers & Fluids*, doi: [10.1016/j.compfluid.2026.107080](https://doi.org/10.1016/j.compfluid.2026.107080).
- 2024 Wave breaking and transport of clifftop boulders, 18th International SPHERIC Workshop proceedings.
- 2023 Implementation of various-fidelity methods for viscous effects modeling on the design of a waverider, *Aerospace Science and Technology*, doi: [10.1016/j.ast.2023.108141](https://doi.org/10.1016/j.ast.2023.108141).

CONFERENCES

- 2025 2nd European Fluid Dynamics Conference, 2025, University College Dublin, Dublin, Ireland.
- 2024 Multiphase, Duisburg, Germany.
- 2024 18th International SPHERIC Workshop, Berlin, Germany.
- 2023 3rd International Workshop on Waves, Storm Surges, and Coastal Hazards Incorporating the 17th International Waves Workshop, University of Notre Dame, Notre Dame, Indiana, USA.
- 2022 National High-Performance Compute and Data Ecosystem Symposium (HiCoDES), Dublin, Ireland.
- 2022 6th DualSPHysics Workshop, Universitat Politècnica de Catalunya, Barcelona, Spain.